BRAIND AI

Senior AI Engineer
Technical Assessment

CONFIDENTIAL

Time Limit: 8 Hours | Submission: Same Day

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Overview

Welcome to the BraindAI Senior AI Engineer Technical Assessment.

This is not a traditional coding test. We are not interested in whether you can solve a LeetCode problem or write a sorting algorithm from scratch. We want to see how you think, how you architect, how fast you move, and how you use modern AI tools to multiply your output.

At BraindAI, we build custom AI automation systems for real businesses. Our engineers take a client brief, decompose the problem, architect a solution, and ship a working system — often within days. This assessment simulates exactly that.

TIME LIMIT
You have 8 hours from the moment you begin. There are no extensions. This constraint is intentional — we want to see how you prioritise, scope, and ship under pressure. Quality matters, but so does knowing when something is good enough to ship.

The Client Brief

Client: SecureLife Insurance Brokers (fictional)

Industry: Insurance

Company size: 45 employees, 3 offices across India

The Problem

SecureLife currently handles new client inquiries manually. A potential customer reaches out (via their website, WhatsApp, or phone), and a sales agent manually collects their information, asks qualifying questions, and enters the data into a spreadsheet. If the prospect is interested, they email over their existing insurance documents (policy PDFs, claim histories, etc.). An analyst then manually reads through these documents, extracts the relevant data points, and creates a summary for the senior broker to review.

This process takes 2–3 days per lead. SecureLife loses prospects to faster competitors. The manual extraction is error-prone. The senior broker wastes time on leads that should have been filtered earlier. There is no centralised place to see the pipeline.

What SecureLife Wants

They want an intelligent system that does the following:

1. **Lead Capture:** An AI-powered conversational interface (chatbot) that engages with prospective clients, asks the right qualifying questions, and captures structured lead data. The chatbot should feel natural, not like a rigid form. It should handle edge cases gracefully — vague answers, incomplete information, follow-up clarifications.
2. **Data Storage:** All captured lead data should be stored in a structured database. Each lead should have a clear status, contact details, qualification scores, and a full conversation history. Think of this as a lightweight CRM purpose-built for their workflow.
3. **Document Processing:** Once a lead is qualified, they upload their existing insurance documents. The system should automatically extract key information from these documents: policy type, coverage amounts, premium details, exclusions, renewal dates, provider information, and any other relevant data points. The extraction must handle real-world messiness — scanned PDFs, inconsistent formats, missing fields.
4. **Intelligent Analysis:** Based on the extracted document data and the lead's profile, the system should generate an analysis: coverage gaps, potential savings, risk flags, and a recommended action for the broker. This is where AI adds real value — not just extraction, but insight.
5. **Dashboard:** A clean, professional frontend where the SecureLife team can see their entire pipeline at a glance: new leads, leads in progress, documents pending review, completed analyses. Individual lead views should show the full journey — from initial conversation through document analysis to final recommendation.

What You Need to Deliver

You have 8 hours. You will not build a production-ready system in that time, and we do not expect you to. What we expect is a clear demonstration of how you think, how you architect, and how you execute. Here is what you must submit:

Deliverable 1: Architecture Document

A written document (Markdown, Notion page, or PDF) that covers:

- System architecture: What are the core components? How do they interact? What does the data flow look like end to end?
- Tech stack choices: What technologies, frameworks, and services would you use for each layer? Why? What trade-offs did you consider?
- Data model: How would you structure the database? What are the key entities, relationships, and fields?
- AI integration strategy: How would you use LLMs in the system? What goes to an API vs. what runs locally? How do you handle prompt engineering, output validation, cost management, and fallback logic?
- Edge cases and failure modes: What could go wrong? How would you handle it? Think about bad document formats, API failures, ambiguous chatbot inputs, rate limits, and data inconsistencies.

> **NOTE**
> This document should read like something you would present to a technical lead or CTO. Clear, structured, opinionated. Not a vague overview — we want to see specifics.

Deliverable 2: System Diagrams

Visual representations of your architecture. At minimum:

- A high-level system architecture diagram showing all components, services, and data flows.
- An entity-relationship diagram for your data model.
- A sequence or flow diagram for at least one critical workflow (e.g., the lead capture to analysis pipeline).

Use whatever tool you are fastest with — Mermaid, Excalidraw, draw.io, Whimsical, Figma, or even AI-generated diagrams. We care about clarity and completeness, not which tool you used.

Deliverable 3: Working MVP

A functional prototype that demonstrates the core pipeline. This does not need to be feature-complete, but it must actually work. Specifically:

- **The chatbot:** A conversational interface that captures lead information through natural dialogue. It should work — you should be able to have a real conversation with it and see structured data come out the other side.
- **The data layer:** A database with a working schema. Leads captured by the chatbot should actually be stored and retrievable.
- **Document processing:** Upload a document (you can use a sample insurance PDF) and have the system extract structured data from it using an LLM. The extraction should be stored against the relevant lead.
- **The dashboard:** A frontend that displays leads, their status, and extracted document data. It should look polished — not a raw HTML dump. Presentation matters.

IMPORTANT
You choose the stack. We are not prescribing specific tools. Use whatever you believe is the best fit for the problem and whatever lets you move the fastest. Your choices are part of the evaluation.

Deliverable 4: Loom Walkthrough

Record a Loom video (or similar screen recording) that gives a complete overview of everything you built. Walk us through your architecture, demo the working system, show how you used AI tools (Claude Code, Cursor, etc.) during the build, and briefly discuss what you would improve with more time. Keep it concise and structured — aim for 10–15 minutes.

WHY LOOM?
BraindAI is an async-first team. Every feature we ship includes a Loom walkthrough for review. Your ability to communicate clearly, concisely, and professionally on video is not optional — it is a core part of the role.

Evaluation Criteria

Your submission will be scored across six dimensions. Here is exactly what we are looking for:

Criteria	Weight	What We Are Looking For
Architecture & System Design	**25%**	Did you decompose the problem well? Is the architecture clear, scalable, and well-reasoned? Are your tech stack choices justified? Did you think about failure modes?
AI Integration & Prompt Engineering	**20%**	How did you use LLMs in the system? Is the prompt design thoughtful? Do you validate outputs? Do you handle edge cases like hallucinations, malformed responses, or API failures?
Execution Speed & AI-Assisted Development	**20%**	How fast did you move? Did you use Claude Code or similar tools effectively? Can you show evidence of AI-accelerated development in your Loom walkthrough?
Working MVP Quality	**15%**	Does it actually work? Can we run it and see the pipeline in action? Is the code clean enough to understand? Not production-grade, but functional and sensible.
Frontend & Presentation	**10%**	Does the dashboard look professional? Is the UI intuitive? Does it feel like something you could show a client, or does it look like a hackathon demo?
Communication & Loom Quality	**10%**	Is the walkthrough clear and well-structured? Can you explain technical decisions to a non-technical audience? Is the video concise and professional?

Rules and Guidelines

What Is Allowed

- Use any programming language, framework, or tool you want. There are no stack restrictions.

- Use AI-assisted coding tools (Claude Code, Cursor, Copilot, etc.). In fact, we expect you to. This is how we work at BraindAI.
- Use open-source libraries, templates, and boilerplate. Do not waste time building what already exists.
- Use any database, hosting platform, or third-party service that helps you move faster.
- Refer to documentation, Stack Overflow, or any other reference material.

What Is Not Allowed

- Submitting pre-built work. Everything must be built during the 8-hour window. We will check timestamps.
- Having another person help you build the solution. This is an individual assessment.
- Copying an existing project and submitting it as your own. Using templates and boilerplate is fine — submitting a pre-existing application is not.

Practical Notes

- You may use free tiers of any cloud services (Supabase free tier, Vercel hobby plan, etc.). Do not spend money on this assessment.
- For document processing, you will need sample insurance documents to test with. Download a few from the links below, or create your own test documents.
- If you hit a genuine blocker (API rate limits, service outage, etc.), document it and show us what you would have built. We understand that 8 hours is tight.
- Focus on depth over breadth. A working chatbot with excellent AI integration and a clean dashboard is better than five half-finished features.

Sample Documents for Testing

Use any of the following resources to download sample insurance documents for your document processing pipeline:

- Insurance Fact Finder Form — secureinsurancegroup.com/wp-content/uploads/2021/10/Fact-Finder-ING.pdf
- Sample Insurance Policy Templates — templateroller.com/tags/109493-insurance-policy/
- Insurance Document PDF Templates — jotform.com/pdf-templates/insurance
- Insurance Claim & Application Forms — sampleforms.com/insurance-claim-form.html

You are free to use any other sample insurance documents you find online. The goal is to have realistic PDFs to demonstrate your extraction pipeline.

Submission Instructions

Submit the following to the email address provided in your assessment invitation:

6. **A GitHub repository** (public or private — if private, invite us) containing all your code, architecture document, diagrams, and a README with clear setup instructions. We should be able to get it running locally in under 5 minutes.
7. **A Loom video link** with a complete walkthrough of everything you built — architecture, demo, AI workflow, and improvements. Include this link in both your README and your submission email.

DEADLINE

All materials must be submitted within 8 hours of receiving this assessment. Late submissions will not be reviewed. The timer starts when you receive this document.

What a Great Submission Looks Like

We have reviewed hundreds of technical assessments. Here is what separates the top candidates from everyone else:

They start with thinking, not coding. The best candidates spend the first 60–90 minutes on architecture and planning. They draw diagrams, define their data model, and map out the pipeline before writing a single line of code. This investment pays off massively in execution speed later.

They use AI tools like a power user, not a beginner. They do not just ask Claude Code to “build me a chatbot.” They break the work into specific, well-scoped prompts. They iterate. They know when to trust the output and when to rewrite. Their Loom video shows a fluent, practiced workflow — not someone using the tool for the first time.

Their MVP actually works. You can clone the repo, follow the README, and have it running in minutes. The chatbot has a real conversation. The document upload extracts real data. The dashboard shows real results. It is not perfect, but it is functional and demonstrable.

The frontend looks like it was built by someone who cares. It does not need to be a design masterpiece, but it should look professional and intentional. Clean typography, sensible layout, consistent spacing. If you would not show it to a client, do not submit it.

They communicate with clarity and confidence. Their Loom walkthrough is structured, concise, and covers all four points listed above. They explain their decisions without rambling. They are honest about shortcuts and trade-offs.

Good luck. Build something great.

— The BraindAI Team